

AIVE: Active Imagined View Exploration for Visual Spatial Reasoning

Anonymous ACL submission

Abstract

While Vision-Language Models (VLMs) have excelled in 2D understanding and shown promise in 3D domains, their spatial reasoning is often hindered by incomplete visual observations. Since active physical exploration to gather missing information is usually impractical, leveraging visual generation to imagine the unobserved views provides a more feasible and efficient solution. We propose AIVE, a novel method that shifts spatial Visual Question Answering (VQA) from passive inference to an active, imagination-augmented reasoning process. Departing from prior approaches that suffer from substantial inference latency, AIVE introduces a highly efficient and precise framework: a VLM Planner generates targeted exploration strategies, seamlessly guided by a generative World Model Dreamer that synthesizes 3D-consistent future views. To facilitate training and evaluation, we introduce SpaThor, a benchmark comprising expert trajectories and action-conditioned visual transitions. Extensive experiments demonstrate that AIVE achieves state-of-the-art performance on challenging 3D spatial reasoning tasks, outperforming prior baselines by an average margin of 5.2%. Notably, it delivers up to a 4.8× inference speedup over existing method based imagination, highlighting its strong potential for real-time embodied applications. All data will be released. An anonymous code repository is available at https://anonymous.4open.science/r/anonymous_code-CAD0/.

1 Introduction

Spatial reasoning serves as a critical capability for embodied agents performing tasks like navigation and object manipulation (Liu et al., 2023; Zhou et al., 2026). Despite their massive success in 2D images, current Vision-Language Models (VLMs) struggle with 3D spatial environments (Zhang et al., 2025). While some methods try to infer spatial relationships from a single static view, they suffer

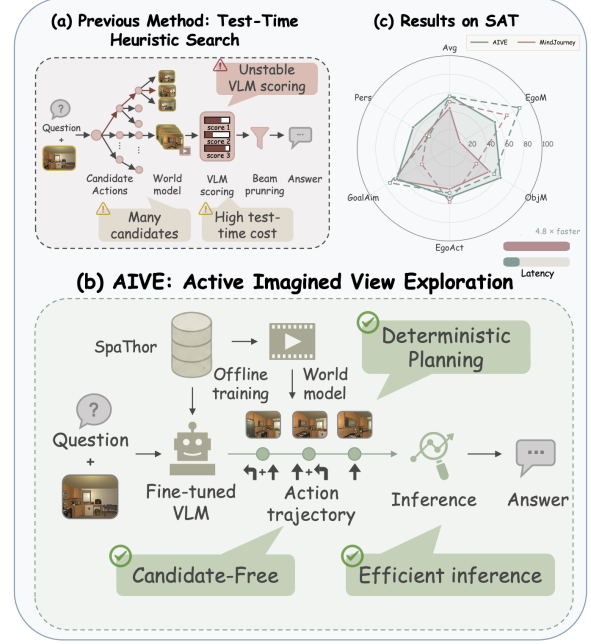


Figure 1: **Conceptual comparison of previous methods and AIVE.** (a) Prior approaches rely on costly test-time heuristic search and noisy VLM scoring over action candidates. (b) In contrast, AIVE directly generates action sequences and informative imagined views, enabling efficient and robust spatial reasoning. (c) Performance comparison on the SAT benchmark.

from a two-fold challenge. Primarily, a single image cannot provide complete spatial context due to incomplete views. Furthermore, typical VLMs do not possess the ability to explicitly model 3D structures from these limited observations (Wasi et al., 2026; Wu et al., 2025).

In the physical world, agents typically resolve spatial ambiguity by moving to gather new informative viewpoints (Bajcsy, 1988; Ramakrishnan et al., 2021). However, executing physical exploration is often impractical. Beyond temporal costs, safety risks, and mechanical latency, systems frequently lack direct control over the agent to perform such movements (Wong et al., 2025). Drawing inspiration from human mental simulation (Hamrick, 2019; Shepard and Metzler, 1971), we observe that

humans naturally bypass physical restrictions by imagining a scene from alternative perspectives. This cognitive process synthesizes necessary spatial evidence without requiring actual physical displacement (Das et al., 2018).

Building on this concept, recent works have begun exploring visual imagination for spatial reasoning by leveraging generative models to synthesize novel views. However, existing imagination-based approaches exhibit critical limitations (Yang et al., 2025c). They typically rely on exhaustive search over action candidates combined with VLM-based scoring for pruning. This paradigm incurs high computational latency and introduces noise through unstable VLM scoring.

To bridge this gap, we propose AIVE, a method that transforms spatial VQA into an active, cognition-inspired process through efficient exploration and imagination. A VLM-based Planner acts as the executive brain to formulate exploration strategies by predicting goal-oriented actions. Concurrently, a generative Dreamer serves as the visual simulator to realize these strategies by synthesizing future views. This synergistic mechanism allows AIVE to explore the environment’s geometry in a closed mental loop, proposing informative viewpoints and generating consistent visual evidence to resolve complex spatial queries without actual physical movement.

To support the training and evaluation of our framework, we construct SpaThor, a novel dataset specifically tailored for active spatial exploration. Training the Planner and Dreamer requires data that explicitly link actions to their resulting visual changes. However, standard VQA datasets rely heavily on static scenes (Koo et al., 2025; Yang et al., 2025b; Ding et al., 2020). To address this gap, we develop an automated pipeline within the AI2-THOR simulator to construct this dataset (Kolve et al., 2017). For training, SpaThor provides precise action trajectories with their corresponding sequential visual transitions. This paired data enables joint supervision of the two components. Furthermore, the SpaThor dataset provides a dedicated test subset, SpaThor-1K, which serves as a challenging benchmark for evaluation. Unlike conventional tasks, it assesses whether a model can resolve spatial ambiguities via dynamic evidence gathering.

Our key contributions are as follows:

- We introduce **AIVE**, a method that shifts spatial reasoning from passive inference to active

evidence gathering. Specifically, it integrates a Planner to formulate exploration strategies and a generative Dreamer to synthesize subsequent views, establishing a closed mental loop to resolve complex spatial queries.

- We construct **SpaThor**, a comprehensive data engine for active spatial reasoning that serves as both a rich source for training and a rigorous benchmark for testing.
- Extensive experiments demonstrate that AIVE achieves state-of-the-art accuracy. Crucially, it delivers a remarkable $4.8\times$ inference speedup over imagination-based baselines, highlighting its viability for real-time embodied applications.

2 Related Work

2.1 Spatial Vision-Language Models

The VLM landscape has evolved from early cross-modal alignment to Large Vision-Language Models (LVLMs) (Li et al., 2025b). Proprietary models such as GPT-4V series and the Gemini series (Yang et al., 2023; Team et al., 2023), alongside open-source frameworks like LLaVA, InternVL, and Qwen-VL (Liu et al., 2023; Wang et al., 2025; Yang et al., 2025a), have achieved near-human performance in general multimodal reasoning.

Despite these advancements, general-purpose LVLMs still fall short in fine-grained spatial reasoning and 3D geometric inference from 2D observations (Pang et al., 2026; Wang et al., 2024). To bridge the gap, recent efforts have sought to enhance spatial intelligence. For instance, Cambrian-S proposes predictive sensing to leverage surprise signals for memory management (Yang et al., 2025b), whereas VLM-3R introduces metric-scale 3D reconstruction to enable precise spatial reasoning from monocular videos (Fan et al., 2025). Complementarily, SpaceMind designs a camera-guided fusion mechanism to explicitly disambiguate geometry by treating viewpoint as an active modality (Zhao et al., 2025).

2.2 World Models

World models provide a cognitive framework for modeling and imagining physical spaces (Wang et al., 2026). By learning the structural dynamics of an environment, they allow agents to mentally simulate future states and visualize unseen areas

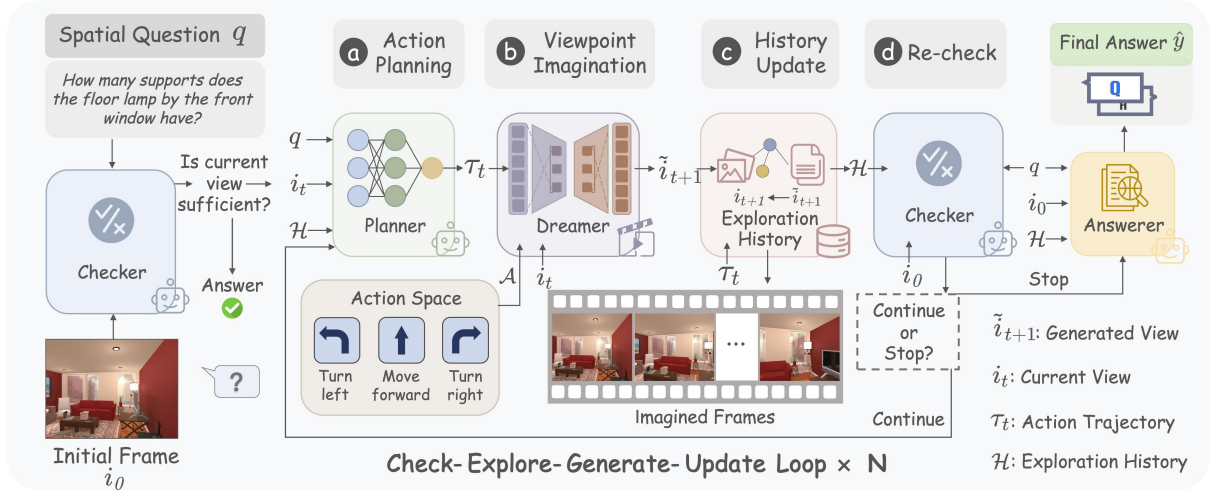


Figure 2: **Overview of the AIVE framework.** If the initial view lacks sufficient information, the Checker triggers the *Active Exploration Loop*, comprising four stages: (a) **Action Planning** via the Planner; (b) **Viewpoint Imagination** using the Dreamer generative model; (c) **History Update** to aggregate generated frames; and (d) **Re-check** to evaluate the updated evidence. Once a STOP signal is emitted or the maximum step limit is reached, the Answerer outputs the final answer.

without requiring actual movement. (Danish et al., 2025; Ding et al., 2025; Xiang et al., 2025). Recent advances have shifted from unconstrained synthesis to controllable generation across three key axes: (1) Camera-conditioned models ensure geometric consistency via precise extrinsics (Xu et al., 2024; Zhou et al., 2025; Yuan et al., 2025); (2) 3D-aware models enforce structural fidelity through geometric priors (Dong et al., 2024; Xia et al., 2026; Wu et al., 2026b); and (3) frameworks driven by semantic prompts synthesize sequential transitions based on text instructions (Wan et al., 2025; Huang et al., 2025). In Embodied AI, such models facilitate data augmentation (Trabucco et al., 2024; Shivashankar and Miller, 2023), reinforcement learning (Matsuo et al., 2022; Wu et al., 2026a; Huang et al., 2024), and imagination-augmented planning.

2.3 Active Vision

Active vision mitigates 3D ambiguities via proactive viewpoint control (Li et al., 2025a). Early efforts focus on active learning for resource optimization, such as reducing computation (Yuan et al., 2023; Chen et al., 2011) or annotation overhead (Griffioen et al., 2024).

More recent works shift toward active viewpoint reasoning. CoV introduces a training-free Chain-of-View framework for multi-step VLM navigation (Zhao et al., 2026), while MindJourney employs an expand-evaluate-prune search coupled with video diffusion for mental imagination (Yang et al., 2025c). However, MindJourney faces two critical bottlenecks: (1) VLM noise, due to unreli-

able scoring from biased or hallucinated backbones; (2) Inference latency, caused by exhaustive search cycles requiring multiple VLM calls per step, leading to prohibitive computational latency.

3 Dataset Construction

3.1 Problem Formulation and Preliminaries

The objective of spatial reasoning is to deduce an accurate semantic answer $\hat{y}$ given a user question q and an egocentric observation i_0 . Since extracting complete 3D relations from a static 2D image is inherently ambiguous, we formulate this task as a problem of active visual exploration, where the agent uncovers latent spatial structures through simulated physical interactions.

We define a discrete action space $\mathcal{A}$ as the set of all possible motion-magnitude pairs:

$$\mathcal{A} = \{(\text{forward}, d) \mid d \in \Delta_{\text{trans}}\} \cup \{(\text{left}, \theta), (\text{right}, \theta) \mid \theta \in \Delta_{\text{rot}}\}, \quad (1)$$

where Δ_{trans} and Δ_{rot} denote the sets of predefined translational and rotational magnitudes, respectively. An exploration trajectory of length K is then defined as a sequence $\tau = (a_1, a_2, \dots, a_K)$, where each $a_k \in \mathcal{A}$.

To simulate the outcomes of these actions without physical interaction, we introduce an action-conditioned World Model Dreamer $\mathcal{W}$. Given a current visual observation i_t and a planned trajectory τ_t , the Dreamer predicts an imagined future view $\tilde{i}_{t+1} = \mathcal{W}(i_t, \tau_t)$. The imagined view $\tilde{i}_{t+1}$

Algorithm 1 AIVE: Active Imagined View Exploration

Require: question q ; initial frame i_0 ; Checker $\mathcal{V}_{\text{check}}$; Planner $\mathcal{V}_{\text{plan}}$; Answerer $\mathcal{V}_{\text{ans}}$; action space $\mathcal{A}$; Dreamer $\mathcal{W}$; maximum exploration steps T

Ensure: Final answer $\hat{y}$ to question q

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1:  $i \leftarrow i_0$ 
2:  $\mathcal{H} \leftarrow \emptyset$ 
3:  $z \leftarrow \mathcal{V}_{\text{check}}(q, i_0)$ 
    $z \in \{\text{EXPLORE}, \text{ANSWER}\}$ 
4: if  $z = \text{ANSWER}$  then
5:   return  $\mathcal{V}_{\text{ans}}(q, i_0)$ 
6: end if
7: for  $t = 0$  to  $T$  do
8:    $\tau_t \leftarrow \mathcal{V}_{\text{plan}}(q, i_t, \mathcal{A}, \mathcal{H})$ 
9:    $\tilde{i}_{t+1} \leftarrow \mathcal{W}(i_t, \tau_t)$ 
10:   $\mathcal{H} \leftarrow \mathcal{H} \cup \{(t, \tau_t, \tilde{i}_{t+1})\}$ 
11:   $i_{t+1} \leftarrow \tilde{i}_{t+1}$ 
12:   $z \leftarrow \mathcal{V}_{\text{check}}(q, i_0, \mathcal{H})$ 
    $z \in \{\text{CONTINUE}, \text{STOP}\}$ 
13:  if  $z = \text{STOP}$  then
14:    break
15:  end if
16: end for
17: return  $\mathcal{V}_{\text{ans}}(q, i_0, \mathcal{H})$ 

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then serves as the initial observation for the subsequent exploration step. Over T exploration steps, the agent incrementally aggregates these sequential states to construct an episodic exploration history $\mathcal{H} = \{(t, \tau_t, \tilde{i}_{t+1})\}_{t=1}^T$.

Finally, a frozen VLM Answerer $\mathcal{V}_{\text{ans}}$ synthesizes the initial observation i_0 , the question q , and the exploration history $\mathcal{H}$ to produce the final answer $\hat{y}$. The complete procedural pipeline is formalized in Algorithm 1.

3.2 Data Format

The AIVE incorporates two learnable modules requiring supervised training data: the Planner $\mathcal{V}_{\text{plan}}$, which learns to predict goal-directed camera trajectories from a question and the current view, and the Dreamer $\mathcal{W}$, which learns to synthesize novel views from an egocentric frame and a camera trajectory. These objectives jointly define each training sample as containing both a spatial question paired with a ground-truth action trajectory and the corresponding visual transitions with precise camera poses. We automatically collect this data using the

AI2-THOR indoor simulator (Kolve et al., 2017), which provides ground-truth geometry, segmentation, and controllable camera motion.

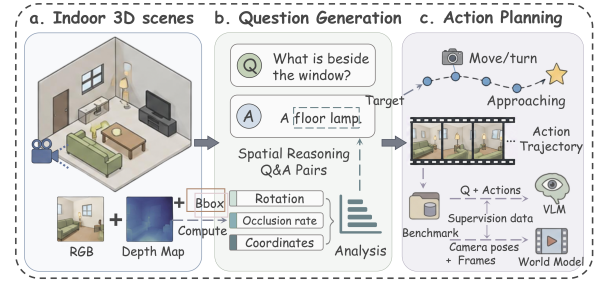


Figure 3: **Construction and Applications of SpaThor.** Built upon AI2-THOR, the dataset is constructed through spatial attribute extraction, automatic QA generation, and ground-truth trajectory collection, supporting both question-conditioned action planning and action-conditioned view synthesis.

Therefore, to meet these requirements, we construct SpaThor, a challenging 3D Q&A dataset tailored for training. Each sample in our dataset consists of two aligned components.

Planning Record. The planning record provides supervision for the Planner $\mathcal{V}_{\text{plan}}$. It is a tuple (q, i_0, τ) , where q is a spatial question, i_0 is the initial egocentric frame, and $\tau = (a_1, \dots, a_K)$ is the ground-truth action trajectory expressed in natural language. This record trains $\mathcal{V}_{\text{plan}}$ to generate, given q and i_0 , an action sequence that navigates toward the spatial evidence needed to answer q .

Trajectory Record. The trajectory record provides supervision for the Dreamer $\mathcal{W}$. It is a tuple $(i_0, \{i_k\}_{k=1}^K, \{P_k\}_{k=1}^K)$, where i_k is the RGB frame observed after executing the k -th action and $P_k \in \mathbb{R}^{3 \times 4}$ is the corresponding camera-to-world transformation matrix in a right-handed, Y-up coordinate frame. Because each discrete physical action a_k deterministically corresponds to a relative spatial transformation ΔP_k , this record trains $\mathcal{W}$ to synthesize the subsequent view $\tilde{i}_k$ based on the geometric mapping $\tilde{i}_k = \mathcal{W}(i_{k-1}, \Delta P_k)$.

3.3 Generation Pipeline

We construct the dataset automatically via a three-stage pipeline built on AI2-THOR see in Figure 3. First, a scene sampler teleports the agent to diverse reachable viewpoints, collecting the initial frame i_0 . An occlusion analyzer leverages instance segmentation and depth maps to quantify object visibility, prioritizing targets that are partially observable and thus suitable for active exploration. Second, a question generator leverages the spatial attributes

extracted in the previous stage to sample from pre-defined spatial question types, constructing QA tuples centered on the selected target. Finally, an action searcher executes a greedy visual-servoing policy from i_0 toward the target, recording natural-language actions, the corresponding ground-truth observations, and ground-truth camera poses to produce the aligned Q&A and trajectory records. Further details regarding the data generation pipeline are provided in the Appendix B.

4 Method

4.1 Overview of AIVE

The core philosophy of Active Imagined View Exploration (AIVE) is to reformulate spatial reasoning from passive inference to an active process of mental simulation. This shift addresses the fundamental limitations of traditional VLMs, which often struggle with 3D queries due to severely limited views, while our efficient and precise action planning ensures strong reasoning performance while substantially mitigating inference latency.

As illustrated in Figure. 2, (i) the workflow initiates with a frozen VLM Checker $\mathcal{V}_{\text{check}}$ evaluating the question q relative to the initial frame i_0 , immediately invoking the Answerer $\mathcal{V}_{\text{ans}}$ if the visual evidence is sufficient for reasoning. (ii) In cases of persistent spatial ambiguity, the system enters an exploration loop where the Planner $\mathcal{V}_{\text{plan}}$ and the Dreamer $\mathcal{W}$ function as the executive and generative components of the framework. Specifically, $\mathcal{V}_{\text{plan}}$ analyzes the current state and exploration history $\mathcal{H}$ to propose a predicted action trajectory τ_t , which the $\mathcal{W}$ uses with i_0 to synthesize a geometrically consistent future view $\tilde{i}_{t+1}$, and $\tilde{i}_{t+1}$ serves as the starting point for the next step $t + 1$. (iii) This cycle of planning and synthesis continues until $\mathcal{V}_{\text{check}}$ determines that the accumulated visual evidence in $\mathcal{H}$ is adequate for termination. (iv) Finally, $\mathcal{V}_{\text{ans}}$ receives the initial observation i_0 and exploration history $\mathcal{H}$ to deduce the final response $\hat{y}$. Further details regarding inference for AIVE are provided in the Appendix A.1.

To instantiate this pipeline, we leverage the SpaThor to bridge the gap between planning and imagination. $\mathcal{V}_{\text{plan}}$ is fine-tuned on expert trajectories to learn effective exploration policies, while the $\mathcal{W}$ is trained to capture the mapping between agent movement and visual transitions.

4.2 Planner: Prompting and Supervised Fine-Tuning

The objective of the Planner $\mathcal{V}_{\text{plan}}$ is to resolve spatial uncertainty by proposing an optimal action sequence. Given the initial view i_0 , the spatial query q , and a task-specific prompt $\mathcal{P}_{\text{plan}}$, $\mathcal{V}_{\text{plan}}$ generates a discrete action trajectory $\tau = \{a_1, a_2, \dots, a_K\}$ to guide the mental simulation.

Zero-shot Prompting Mechanism. We first establish a zero-shot baseline where a VLM generates an exploration trajectory $\tau = \mathcal{V}_{\text{plan}}(i_0, q, \mathcal{P}_{\text{plan}})$. Despite rich semantic priors, this approach lacks the embodied intuition required for 3D navigation. As a result, it often produces sub-optimal or inconsistent trajectories in complex environments.

Supervised Fine-Tuning with Expert Trajectories. To bridge the gap between semantic understanding and embodied spatial planning, we explicitly fine-tune $\mathcal{V}_{\text{plan}}$. We formulate this as a multimodal autoregressive sequence generation task. Leveraging the aforementioned Planning Records from our dataset SpaThor, we supervise the model with expert-crafted optimal trajectories:

$$\mathcal{L}_{\text{plan}} = - \sum_{j=1}^k \log p(a_j^* | i_0, q, a_{<j}^*, \mathcal{P}_{\text{plan}}), \quad (2)$$

where a_j^* represents the ground-truth action. Through this targeted optimization, $\mathcal{V}_{\text{plan}}$ transcends the uninformed semantic heuristics of the zero-shot and internalizes the underlying geometric priors of 3D environments.

4.3 Dreamer: Action-Conditioned Imagined View Generation

The Dreamer $\mathcal{W}$ serves as a mental simulator to hallucinate future states. We employ controllable video generation models, specifically Wan2.2-TI2V-5B (Wan et al., 2025) as the backbone of $\mathcal{W}$. We train $\mathcal{W}$ on trajectory records by extracting consecutive frames (i_{t-1}, i_t) and computing the relative camera pose $\Delta P_t = P_{t-1}^{-1} P_t \in SE(3)$. The model is optimized via the denoising loss:

$$\mathcal{L}_{\text{world}} = \mathbb{E} \left\| \epsilon - \epsilon_{\theta}(z_t^{(k)}, k, z_{t-1}, \Delta P_t) \right\|_2^2, \quad (3)$$

where z_{t-1} and ΔP_t provide the visual context and action condition, respectively. During inference, $\mathcal{W}$ synthesizes the imagined next view $\tilde{i}_{t+1}$ to provide visual evidence for spatial deduction.

Type	# Samples	Target Ability
Distance	300	Distance perception and metric estimation.
Occlusion	247	Occlusion reasoning under partial visibility.
Spatial Relation	453	Directional and topological relation reasoning.
Total	1,000	Comprehensive spatial reasoning evaluation.

Table 1: Statistics of the SpaThor-1K benchmark.

4.4 Evidence Integration and Reasoning

The Checker $\mathcal{V}_{\text{check}}$ and the Answerer $\mathcal{V}_{\text{ans}}$ perform zero-shot reasoning over the exploration history $\mathcal{H} = \{(t, \tau_t, \tilde{i}_t)\}$, bridging discrete imagined steps with high-level semantic deduction.

Dynamic Checking. $\mathcal{V}_{\text{check}}$ monitors the sequence of synthesized views against the spatial query q . Acting as a semantic gatekeeper, it triggers a STOP signal upon identifying sufficient target cues, effectively mitigating hallucination drift caused by excessive exploration.

History Aggregation. Upon termination, $\mathcal{V}_{\text{ans}}$ aggregates the initial observation i_0 , the executed action sequences $\{\tau_t\}$, and all imagined views $\{\tilde{i}_t\}$ into a unified visual-spatial chain. By reasoning over this enriched context, $\mathcal{V}_{\text{ans}}$ deduces the final answer $\hat{y}$, thereby closing the AIVE loop through holistic evidence integration.

5 Experiments

5.1 Experimental Setup

Benchmark. We evaluate our model on two benchmarks: the SpaThor-1K test set, derived from our proposed SpaThor dataset, and the existing SAT benchmark (Ray et al., 2024).

We primarily evaluate our model on SpaThor-1K, a challenging benchmark we proposed to assess embodied spatial reasoning. As summarized in table 1, SpaThor-1K is constructed within the AI2-THOR simulator and consists of 1,000 Q&A pairs. These cases are stratified into three categories: distance, occlusion, and spatial relationship. Further details regarding SpaThor-1K are provided in the Appendix 15.

To further verify the model’s fundamental spatial cognition, we also report results on the SAT benchmark. This includes SAT-Synthesized, a large-scale simulated set of 4,000 Q&A pairs, and SAT-Real, which comprises 150 challenging real-world pairs

captured across diverse indoor and unstructured outdoor environments.

Baselines. We compare AIVE against two lines of baselines. (i) Direct inference: we evaluate direct spatial reasoning capabilities using a diverse suite of SOTA VLMs, including closed-source models such as GPT-4o and GPT-5 (Hurst et al., 2024; Singh et al., 2025), as well as high-performance open-source models such as InternVL-3 and Qwen3 (Wang et al., 2025; Yang et al., 2025a). (ii) Active vision pipelines: we benchmark against recent SOTA imagination-based and active exploration methods, with MindJourney serving as our primary baseline (Yang et al., 2025c).

Evaluation Metrics. We evaluate reasoning performance using standard Accuracy. To assess practicality for real-time embodied applications, we further report inference latency and compare it against existing imagination-based baselines.

Implementation Details. During inference, we set the maximum exploration steps to $T = 3$. To enable precise manipulation, we define a fine-grained action space: translation is quantized to $0.25 \times N$ meters up to a maximum of 3m, and rotation is quantized to $9^\circ \times N$ up to a limit of 90° . Our Planner and Dreamer are initialized from the pre-trained InternVL-8B (Wang et al., 2025) and Wan2.2-TI2V-5B (Wan et al., 2025), respectively. Comprehensive training details are provided in the Appendix A. All experiments, including training and inference, are conducted on NVIDIA A100 GPUs.

5.2 Main Results

We evaluate AIVE on the public SAT benchmark and our proposed SpaThor-1K dataset. While SAT offers a standard testbed, SpaThor-1K emphasizes challenging embodied scenarios requiring active reasoning under limited observability.

Results on SAT. As shown in Table 2, AIVE surpasses all baselines on both splits. On SAT-Real, it achieves average gains of 7.1% over static and 4.7% over MindJourney, with the largest boost in EgoM. This underscores the effectiveness of AIVE in resolving egocentric ambiguity via geometrically consistent mental simulation. Strong performance on SAT-Synthesized confirms that AIVE generalizes well across diverse simulated environments.

Results on SpaThor-1K. Table 3 shows that AIVE yields a 4.9% average improvement over static and +4.4% over MindJourney across all spatial categories. Unlike MindJourney’s heuristic exploration, AIVE’s learned planning acquires more

Table 2: Accuracy comparison on SAT-Real and SAT-Synthesized. MJ denotes MindJourney. Note that bold numbers indicate the best performance.

Backbone	Method	SAT-Real						SAT-Synthesized					
		Avg	EgoM	ObjM	EgoAct	GoalAim	Pers	Avg	EgoM	ObjM	EgoAct	GoalAim	Pers
GPT-4o (Hurst et al., 2024)	Baseline	59.3	65.2	68.1	55.8	50.0	63.6	61.8	91.1	59.6	50.0	75.1	43.1
	MJ	61.3	60.8	68.1	61.7	63.1	54.5	71.1	88.1	88.7	57.5	95.5	41.2
	AIVE	62.0	60.9	63.6	63.2	55.3	72.7	74.3	88.1	78.3	64.7	87.6	49.4
GPT-5 (Singh et al., 2025)	Baseline	74.6	100	63.6	73.5	84.2	54.5	70.6	87.7	89.3	57.1	95.1	39.4
	MJ	77.3	100	82.6	91.2	83.8	36.4	72.3	90.2	77.3	68.2	90.5	39.4
	AIVE	80.6	100	86.4	88.2	86.8	48.4	74.7	90.2	82.4	70.6	87.6	41.5
InternVL3-8B (Wang et al., 2025)	Baseline	40.6	0.0	47.8	47.0	62.0	33.0	47.8	78.8	45.9	49.0	43.5	25.8
	MJ	43.3	13.0	50.0	44.1	68.4	30.3	50.3	71.4	33.3	57.8	34.7	28.3
	AIVE	56.7	56.5	63.6	52.9	65.8	45.5	56.2	87.4	55.7	48.3	74.5	25.8
Qwen3-8B (Yang et al., 2025a)	Baseline	49.0	65.2	43.4	67.6	42.1	30.3	47.9	80.0	53.9	23.3	85.1	17.5
	MJ	51.3	60.8	45.5	73.5	50.0	27.2	45.8	80.0	53.2	24.4	76.1	14.1
	AIVE	52.6	73.9	43.4	79.4	55.2	33.3	48.8	80.0	53.2	27.4	85.1	25.8

Table 3: Accuracy comparison on SpaThor-1K. MJ denotes MindJourney, and AIVE denotes our proposed augmentation strategy. Note that bold numbers indicate the best performance.

Backbone	Method	SpaThor-1K			
		Avg.	Dist.	Occ.	SpatRel.
GPT-4o	Baseline	57.2	55.4	44.8	65.1
	MJ	57.2	56.5	47.9	62.8
	AIVE	60.7	61.8	52.1	64.6
GPT-5	Baseline	65.3	53.4	56.7	77.9
	MJ	68.9	60.5	54.7	82.2
	AIVE	72.2	56.3	61.8	88.3
InternVL3-8B	Baseline	50.8	44.7	40.8	60.3
	MJ	51.7	52.6	40.8	57.1
	AIVE	57.4	55.3	48.9	63.5
Qwen3-8B	Baseline	57.5	48.3	50.6	67.4
	MJ	55.2	39.8	48.3	69.2
	AIVE	60.1	49.3	55.5	69.8

task-relevant observations while avoiding redundant or noisy inputs, proving critical for complex embodied reasoning.

Efficiency Analysis. As shown in Fig. 4, AIVE achieves superior computational efficiency, reducing inference latency by an average of 40.9% compared to MindJourney, with the gap widening significantly as exploration steps increase. This efficiency is attributed to AIVE’s streamlined mental simulation pipeline, which eliminates the redundant computations inherent in heuristic search. Notably, it delivers up to a 4.8× inference speedup over MindJourney, making it well-suited for resource-constrained embodied platforms.

5.3 Ablation Study

Ablation of the Planner. We ablate the Planner by comparing our SFT-based variant against a prompt-

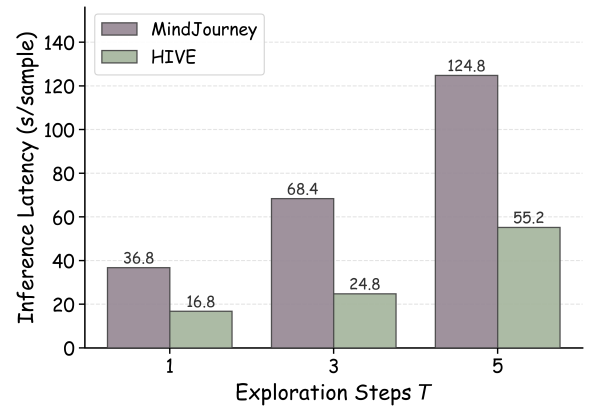


Figure 4: Evaluate the inference latency of AIVE in comparison with MindJourney on SpaThor-1K.

based baseline, keeping all other AIVE components fixed. As shown in Table 4, SFT surpasses prompting across nearly all tasks, yielding up to a 4.8% accuracy gain. While prompting suggests plausible movements, it lacks the task logic to select actions that optimally resolve spatial ambiguities. In contrast, SFT learns strategic exploration patterns that prioritize informative viewpoints and reduce redundancy. This confirms that AIVE’s gains arise from the quality of visual evidence rather than input quantity, establishing specialized planning as a cornerstone for robust embodied reasoning.

Ablation of the Dreamer. We ablate the Dreamer by comparing three variants: a camera-control baseline (SEVA) (Zhou et al., 2025), a prompt-based WAN, and our supervised fine-tuned WAN, keeping all other AIVE components fixed. As shown in Table 5, the Dreamer achieves the highest accuracy 57.4%, outperforming the prompt-based baseline by 3.4% and surpassing SEVA overall. This indicates that training on simulator trajectories yields a more robust mapping between actions and 3D viewpoint changes.

Table 4: Ablation of the Planner on SpaThor-1K. Note that bold numbers indicate the best performance.

Backbone	Method	SpaThor-1K			
		Avg.	Dist.	Occ.	SpatRel.
InternVL3-8B	Prompt-based	52.6	40.8	50.0	61.9
	SFT-based	57.4	55.3	48.9	63.5
Qwen3-8B	Prompt-based	59.3	49.3	51.6	70.2
	SFT-based	60.1	49.3	55.5	69.8

Table 5: Ablation of the Dreamer on SpaThor-1K, the VLM backbone is InternVL3-8B. Note that bold numbers indicate the best performance.

Backbone	Method	SpaThor-1K			
		Avg.	Dist.	Occ.	SpatRel.
SEVA	Camera control	57.1	60.5	46.9	60.3
WAN	Prompt-based	54.0	52.6	46.9	58.7
	SFT-based	57.4	55.3	48.9	63.5

Ablation of Action Planning. To verify the importance of effective action planning, we compare fine-tuned Planner against a Random Walk baseline and Ground-Truth expert trajectories. As shown in Table 6, AIVE outperforms Random Walk by 5.7%. This significant gap confirms that merely increasing visual coverage is insufficient for complex spatial reasoning. AIVE performs within 0.7% of ground-truth trajectories and closely approximates expert evidence gathering, showing that the Planner has effectively captured 3D environmental logic.

Qualitative Results. Figure 5 presents three qualitative examples from the SpaThor-1K dataset. For each example, the left panel displays the Q&A pair, while the right panel illustrates the initial observation and a sequence of future views synthesized by the Dreamer, conditioned on specific action trajectories. Within the images, red and green bounding boxes highlight the anchor and target objects, respectively. As demonstrated, the Dreamer is capable of generating relatively stable and 3D-consistent visual transitions based on camera movements. These synthesized frames effectively provide reliable visual cues to support spatial reasoning during exploration.

However, the view synthesis process also exhibits certain limitations. We observe that while prominent objects occupying larger pixel areas remain structurally stable during camera movements, smaller objects occasionally suffer from spatial drift, distortion, or even complete disappearance. This indicates that maintaining fine-grained geo-

Table 6: Analysis of action strategy. Note that bold numbers indicate the best performance.

Backbone	Method	SpaThor-1K			
		Avg.	Dist.	Occ.	SpatRel.
InternVL3-8B	Random walk	51.7	50.0	40.8	58.7
	Planner	57.4	55.3	48.9	63.5
	Ground-truth	58.1	56.5	48.8	64.2

metric stability for small objects remains a challenging aspect of the current generation process.

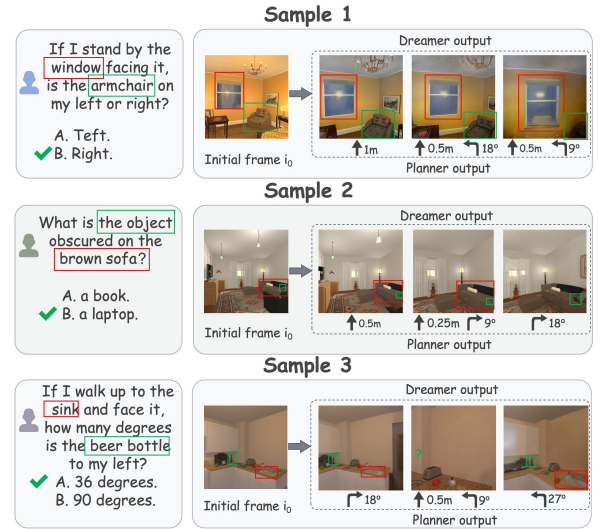


Figure 5: Qualitative Results of AIVE on SpaThor-1K.

6 Conclusion

We present AIVE, a framework that fundamentally reframes spatial visual reasoning from passive observation to active mental simulation. By seamlessly coordinating a goal-directed Planner and an action-conditioned Dreamer, AIVE dynamically resolves egocentric visual ambiguities without requiring physical environment interaction. To facilitate this paradigm, we developed an automated data generation pipeline to construct and contribute the SpaThor dataset. This comprehensive dataset not only provides precisely aligned records for the supervised fine-tuning of both the Planner and the Dreamer, but also introduces a highly challenging evaluation benchmark, SpaThor-1K. Extensive empirical evaluations on SpaThor-1K and the SAT benchmark confirm that AIVE consistently enhances the spatial understanding capabilities of diverse off-the-shelf VLMs, establishing a scalable and robust foundation for future embodied reasoning systems.

Limitations

AIVE has several limitations that point to future directions. (i) Our current framework relies on imagination conditioned on isolated single images. However, real-world embodied scenarios typically involve complex, long-horizon video streams. In such dynamic environments, imagination must be anchored to specific frames within a continuous temporal context rather than relying on static snapshots. Extending AIVE to handle temporal video inputs is a crucial next step. (ii) The generative process exhibits inherent instabilities; fine-grained details or small objects occasionally suffer from spatial drift or structural distortion during view synthesis. Furthermore, generation quality is heavily bottlenecked by the underlying backbone. Exploring task-specific optimization strategies that aim to improve robustness and reduce reliance on the backbone’s raw priors could significantly enhance 3D consistency. (iii) While AIVE introduces an explicit active exploration mechanism for spatial reasoning, the underlying VLMs lack tailored optimization for 3D understanding. Our current scope focuses on the imagination pipeline, leaving the intrinsic spatial perception of the VLM unoptimized. Jointly fine-tuning the foundational VLM with explicit 3D scene priors remains a promising avenue to fundamentally overcome spatial-logic limitations.

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A Implementation Details

A.1 Planner: Autoregressive LoRA Fine-Tuning

Base Models. To demonstrate the generalizability of our framework, the Planner is instantiated using two state-of-the-art 8B-parameter multimodal large language models: InternVL-8B and Qwen3-VL-8B. Both models possess robust visual grounding and spatial comprehension capabilities. We fine-tune them using LoRA, keeping the base model parameters strictly frozen and applying optimization solely to the injected low-rank matrices.

Task Formulation and Objective. The Planner is trained within a multimodal autoregressive sequence generation framework. Conditioned on the initial egocentric observation i_0 and a spatial question q , $\mathcal{V}_{\text{plan}}$ generates a goal-directed action trajectory $\tau = (a_1, a_2, \dots, a_K)$ that progressively navigates toward informative viewpoints.

Each action a_k is expressed in natural language and sampled from a discrete action space $\mathcal{A}$. This space includes translational actions of the form move-forward d with step sizes $d \in \Delta_{\text{trans}}$, and rotational actions turn-left θ and turn-right θ with angles $\theta \in \Delta_{\text{rot}}$. To ensure environment compatibility, translations are quantized to multiples of 0.25 m up to 3 m, and rotations are quantized to multiples of 9° up to 90° .

The training objective follows standard causal language modeling over the action tokens:

$$\mathcal{L}_{\text{plan}} = - \sum_{j=1}^K \log p(a_j^* | i_0, q, a_{<j}^*, \mathcal{P}_{\text{plan}}), \quad (4)$$

where a_j^* represents the j -th ground-truth action in the expert trajectory, $a_{<j}^*$ denotes all preceding actions, and $\mathcal{P}_{\text{plan}}$ is the task-specific system prompt instructing the model to generate 3D-consistent exploration sequences (refer to Appendix D for the exact prompt templates).

Training Data and Input/Output Format. We train the Planner on the Planning Records derived from our SpaThor dataset. Each record is formalized as a tuple (q, i_0, τ) , representing a spatial question, the initial egocentric frame captured in the AI2-THOR simulator, and the corresponding ground-truth action trajectory τ collected by a greedy visual-servoing expert agent.

The full training corpus comprises **100,000** planning records spanning diverse indoor environments

across three core spatial reasoning categories: Distance, Occlusion, and Spatial Relation. The data is formatted in ShareGPT style, structured as a single-turn multimodal conversation illustrated in Table 7, with 10% randomly reserved as a held-out validation split.

Table 7: Input and output format for the Planner training.

Role	Content
User	<image> + spatial question q + prompt
Assistant	turn-left 36 degrees, move-forward 0.75 meters

Training Setup and Hyperparameters. LoRA adapters are applied to all attention and feed-forward layers to ensure sufficient representational capacity. We optimize the model using AdamW with a cosine learning rate schedule and a short linear warm-up. Training is conducted in bfloat16 mixed precision to balance memory efficiency and numerical stability. All detailed hyperparameters are listed in Table 8.

Table 8: Training hyperparameters for the Planner.

Hyperparameter	Value
Validation split	10%
Training epochs	3
LoRA rank r	64
LoRA α	128
LoRA dropout	0.05
Peak learning rate	1.5×10^{-5}
Warmup ratio	0.05
Per-GPU batch size	2
Grad. accumulation	8
Effective batch size	16
Max seq. length	2,048 tokens
Hardware	NVIDIA A100 (80 GB)

Inference Details. At test time, given the initial observation i_0 and the spatial question q , the Planner performs a single autoregressive forward pass with greedy decoding and temperature set to zero, deterministically generating the action trajectory at each exploration step. To avoid infinite loops, the total number of macro-exploration cycles is strictly limited to $T = 3$, as detailed in Section 5.1.

A.2 Dreamer: Action-Conditioned Flow Matching

Base Architecture. We adopt Wan2.2-TI2V-5B as our generative backbone for the Dreamer $\mathcal{W}$. To enable action conditioning, the backbone parameters are kept strictly frozen, and we introduce trainable linear camera encoders and attention projectors into each block. This modification yields approximately 284M trainable parameters, representing roughly 5% of the full network.

Training Data and Pre-computation. Model training utilizes the trajectory records from the SpaThor dataset. The dataset consists of 100,000 expert indoor navigation trajectories, partitioned into 95,000 training and 5,000 validation samples. Each trajectory contains $K = 4$ sequential egocentric frames, along with their corresponding relative camera-to-world transformations in $SE(3)$, providing strong geometric supervision for viewpoint synthesis.

To improve training efficiency, we pre-compute and cache static components. The indoor navigation prompt is encoded by UMT5-XXL into a 4096-dimensional embedding, allowing the text encoder to be fully offloaded. Similarly, to maintain strict temporal independence, each frame is independently encoded by the VAE and cached as latents prior to flow matching.

Core Flow Matching Objective. The Dreamer is trained within the flow matching framework. For a clean latent z_0 and noise $\varepsilon \sim \mathcal{N}(0, I)$, the noisy latent at time $t \in [0, 1]$ is obtained via linear interpolation:

$$z_t = (1 - t) z_0 + t \varepsilon. \quad (5)$$

The network is optimized to predict the flow velocity $v = \varepsilon - z_0$. The training objective minimizes the mean squared error between the predicted and target velocities, conditioned on the camera poses C and text embeddings T :

$$\mathcal{L}_{\text{world}} = \mathbb{E}_{t, z_0, \varepsilon} \|v_{\theta}(z_t, t, C, T) - v\|^2. \quad (6)$$

Hardware and Hyperparameters. All training workloads are distributed across a compute node equipped with four NVIDIA A100 GPUs. The trainable modules are optimized using the AdamW optimizer with automatic mixed precision bfloat16. The detailed hyperparameter configuration used during the flow matching training is summarized in Table 9.

Table 9: Hyperparameter settings used for Dreamer training.

Hyperparameter	Value
Learning rate	3×10^{-6}
Adam betas	(0.9, 0.99)
Weight decay	10^{-2}
Gradient clipping (max norm)	1.0
Global batch size	24
Total training steps	5,000
Warmup steps	100
LR schedule	Constant
Mixed precision	bfloat16

B Details of Dataset Construction

B.1 Environment and Embodied Setup

We build our embodied QA framework upon AI2-THOR, a photorealistic interactive indoor simulator that provides physics-based rendering, deterministic agent control, and ground-truth access to per-object 3D metadata, including spatial coordinates, instance segmentation masks, and depth maps. These properties are essential for constructing spatially grounded QA pairs paired with valid action sequences, which are prohibitively difficult to obtain via real-world data collection.

We utilize 120 diverse indoor scenes spanning four room categories: kitchens, living rooms, bedrooms, and bathrooms. As illustrated in Table 15, these environments provide a rich distribution of spatial layouts, object configurations, and lighting conditions.

B.2 Target Identification and QA Generation

To ensure that the generated spatial questions strictly necessitate embodied exploration, we programmatically identify target objects that are partially observable from the initial viewpoint due to occlusion or image boundary truncation.

Occlusion Analysis. Given a rendered RGB image $I \in \mathbb{R}^{H \times W \times 3}$, a paired depth map $D \in \mathbb{R}^{H \times W}$, and an instance segmentation mask $S \in \mathbb{R}^{H \times W \times 3}$, we first isolate the binary mask for a specific object o :

$$M_o[i, j] = \mathbb{I}[S[i, j] = c_o], \quad (7)$$

where c_o is the unique color assigned to o in S , and $\mathbb{I}[\cdot]$ is the indicator function. The visibility and

spatial location of o are then characterized by:

$$N_o^{\text{vis}} = \sum_{i,j} M_o[i,j], \quad (8)$$

$$d_o^{\text{med}} = \text{median}(\{D[i,j] \mid M_o[i,j] = 1\}). \quad (9)$$

Here, N_o^{vis} denotes the number of visible pixels, and d_o^{med} is the median depth computed over the valid mask region of o .

To determine occlusion, we identify foreground pixels within the object’s vicinity. Let $\Omega_o = \{(i,j) \in B_o \mid D[i,j] < d_o^{\text{med}} - \tau, M_o[i,j] = 0\}$, then:

$$N_o^{\text{occ}} = |\Omega_o|, \quad (10)$$

where B_o represents the axis-aligned bounding box of M_o . The term N_o^{occ} counts the non-object pixels ($\neg M_o[i,j]$) within B_o that are closer to the camera than d_o^{med} , with a tolerance margin $\tau = 0.05$ m applied to suppress depth noise. The occlusion rate is subsequently computed as:

$$\text{OCC}(o) = \frac{N_o^{\text{occ}}}{N_o^{\text{occ}} + N_o^{\text{vis}}}. \quad (11)$$

This metric quantifies the fraction of the object’s projection that is occluded. We explicitly exclude uninformative objects containing fewer than 200 visible pixels, as well as large structural elements exceeding a 3D height of 1.8 m.

Truncation Analysis. Complementary to occlusion, truncation measures the fraction of an object extending beyond the image dimensions $W \times H$. For an object o with bounding box dimensions $w \times h$, let Δx and Δy denote the lengths protruding outside the horizontal and vertical image boundaries, respectively. The truncation rate is simply defined as:

$$\text{TRUNC}(o) = \max\left(\frac{\Delta x}{w}, \frac{\Delta y}{h}\right). \quad (12)$$

Procedural QA Generation. A strictly positive truncation or occlusion rate indicates an incomplete visual state, acting as a gating signal to identify targets requiring active camera movement. Using these partially visible objects, we procedurally generate QA pairs across the three core spatial reasoning categories defined in our dataset: (i) Distance: formulated using the exact 3D spatial coordinates and ground-truth distance metrics provided by the AI2-THOR simulator; (ii) Occlusion: constructed directly based on our dynamically computed occlusion and truncation rates; and (iii) Spatial Relationship: derived from the relative 3D positional

Algorithm 2 Evidence-Driven Action Planning

Require: Initial state s_0 , target object o , topometric graph $\mathcal{G}$, weights λ , maximum servo steps k_{max}

Ensure: Exploration trajectory τ^*

Phase 1: Global Viewpoint Optimization

- 1: Extract valid viewpoints $\mathcal{V}_{\text{valid}} \subseteq \mathcal{G}$ for o
- 2: Select the optimal intermediate viewpoint:

$$v^* \leftarrow \arg \max_{v \in \mathcal{V}_{\text{valid}}} \mathcal{J}(v),$$

$$\mathcal{J}(v) = \lambda_{\text{vis}} \mathbb{E}[\text{Vis}(v, o)] - \lambda_{\text{nav}} C_{\text{path}}(s_0, v) - \lambda_{\text{dist}} D(v, o).$$

- 3: $\tau_{\text{macro}} \leftarrow \text{MotionPlanner}(\mathcal{G}, s_0, v^*)$

- 4: $s_t \leftarrow \text{Execute}(\tau_{\text{macro}})$

Phase 2: Active Visual Servoing

- 5: $\tau_{\text{micro}} \leftarrow \emptyset, k \leftarrow 0$

- 6: **while** $k < k_{\text{max}}$ **do**

- 7: $\mathbf{e}_t \leftarrow \text{CenteringError}(s_t, o)$

- 8: **if** $\|\mathbf{e}_t\|_2 < \epsilon$ **then**

- 9: **break** ▷ Object is visually centered

- 10: **end if**

- 11: $a_t \leftarrow \pi_{\text{servo}}(\mathbf{e}_t)$

- 12: $s_t \leftarrow \text{Execute}(a_t)$

- 13: $\tau_{\text{micro}} \leftarrow \tau_{\text{micro}} \oplus \langle a_t \rangle$

- 14: $k \leftarrow k + 1$

- 15: **end while**

- 16: **return** $\tau^* \leftarrow \tau_{\text{macro}} \oplus \tau_{\text{micro}}$
-

metadata and bounding box configurations of multiple objects within the environment.

B.3 Action Planning and Trajectory Collection

For each selected target, we generate an exploration trajectory τ^* to progressively resolve occlusion and accumulate sufficient visual evidence. As shown in Algorithm 2, the planning process consists of two stages: macro-exploration and micro-adjustment.

In the macro-exploration stage, the agent selects a global viewpoint v^* from the topo-metric graph $\mathcal{G}$ by maximizing a heuristic objective $\mathcal{J}(v)$, which balances expected target visibility, navigation cost, and spatial distance. A motion planner then produces a collision-free macro-trajectory τ_{macro} to reach v^* .

In the micro-adjustment stage, the agent further refines its observation through active visual servoing. Specifically, it computes a centering error $\mathbf{e}_t$ to measure the target’s visual misalignment and applies a servoing policy π_{servo} to per-

form local adjustments until the error falls below a tolerance ϵ . The resulting action sequence forms τ_{micro} , and the final exploration trajectory is given by $\tau^* = \tau_{\text{macro}} \oplus \tau_{\text{micro}}$.

Following the generation of τ^* , the agent executes these actions within the simulator to collect empirical data. At each discrete step k , the environment renders a high-fidelity RGB observation i_k and records the corresponding camera-to-world transformation matrix $P_k \in \mathbb{R}^{3 \times 4}$. Because each action $a_k \in \tau^*$ deterministically induces a relative spatial transformation ΔP_k , this continuous embodied experience serves a dual supervisory purpose. First, the tuple (i_0, q, τ^*) trains the Planner to learn a proactive exploration policy from incomplete initial cues. Second, the multi-view visual sequence $(i_0, \{i_k\}_{k=1}^K, \{P_k\}_{k=1}^K)$ acts as ground-truth spatial data to fine-tune $\mathcal{W}$, enabling it to learn 3D geometry and synthesize forward visual predictions via $\tilde{i}_k = \mathcal{W}(i_{k-1}, \Delta P_k)$.

C Additional Experiments

C.1 Impact of Maximum Exploration Steps T

We investigate the impact of the maximum exploration steps T , which controls the upper bound of action-planning loops. When $T = 0$, AIVE performs no imagined exploration and directly answers based on the initial observation. For $T > 0$, AIVE iteratively plans actions, generates imagined views, and uses $\mathcal{V}_{\text{check}}$ to decide whether to continue or stop. Thus, the actual number of executed steps may be strictly smaller than T .

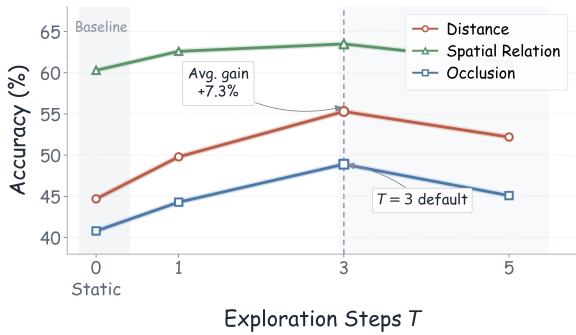


Figure 6: Effect of maximum exploration steps T on the SpaThor-1K.

As shown in Figure 6, increasing T consistently improves performance over the no-exploration setting ($T = 0$), confirming that active evidence acquisition is crucial for spatial reasoning under incomplete observations. This gain is especially pronounced for occlusion and spatial relation questions, which often require visual information be-

yond the initial view. However, the performance improvement gradually saturates as T grows larger, suggesting that most necessary evidence can be acquired within a few exploration steps. We therefore set $T = 3$ as the default in our main experiments, which achieves a favorable trade-off between reasoning accuracy and computational efficiency.

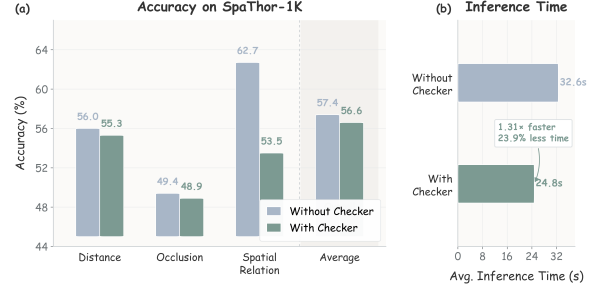


Figure 7: Effect of Exploration Gate on SpaThor-1K.

C.2 Effect of the Exploration Gate

We study the effect of the exploration gate $\mathcal{V}_{\text{check}}$ in AIVE. To avoid coupling this analysis with the exploration budget, we fix the maximum exploration steps T and only vary whether the checker is enabled. The variant without the checker removes both the initial checking and early-stop decisions, forcing the model to exhaust the fixed exploration budget before answering. The full model, in contrast, can directly answer from the initial view or stop exploration early once sufficient evidence has been collected.

Figure 7 reports the accuracy and average inference time on the SpaThor dataset. Without the checker, the model may obtain slightly higher accuracy by passively aggregating more imagined observations, but it incurs a much higher inference time. With the checker enabled, AIVE achieves comparable accuracy at a significantly lower computational cost, indicating that many queries do not require full exploration. This confirms that the exploration gate is highly effective for adaptive and efficient spatial reasoning.

D Prompt Templates

In this section, we present the prompt templates used by our framework to guide the VLMs, namely the Checker (Table 10 and Table 11), the Planner (Table 12), and the Answerer (Tables 13 and 14).

Table 10: Prompt for the Checker (Initial Check)

Prompt for the Checker (Initial Check)
System Prompt: You are an AI assistant designed to help with spatial reasoning in a 3D scene. Your task is to analyze the provided image(s) and a spatial reasoning question, then determine if you need to move the camera to explore the scene further. Guidelines: 1. Carefully observe what is visible in the current image(s). 2. Consider if the question can be answered based solely on what you can see now. 3. If you need to see something that is not currently visible (e.g., behind objects, around corners, different angles, objects that are partially occluded), then exploration IS needed. 4. If all relevant information is already visible or can be inferred from the image(s), exploration is NOT needed. CRITICAL: Output EXACTLY one word — either EXPLORE or ANSWER. No dashes, no bullets, no punctuation, no explanation. Just the single word.

Table 11: Prompt for the Checker (In-Loop Evaluation)

Prompt for the Checker (In-Loop Evaluation)
System Prompt: You are an AI assistant helping with spatial reasoning. You have been exploring a 3D scene by taking actions and observing the results. Based on the sequence of images you've seen and the original question, decide whether you need to continue exploring. Decision Rules: 1. OUTPUT STOP IF: The information currently available (from all images seen so far) is sufficient to answer the question with confidence. 2. OUTPUT CONTINUE IF: Critical information is still missing, obscured, or requires a different viewpoint. CRITICAL: Output EXACTLY one word — either STOP or CONTINUE. No dashes, no bullets, no punctuation, no explanation. Just the single word.

Table 12: Prompt for the Planner

Prompt for the Planner
System Prompt: You are an AI assistant designed to help with spatial reasoning in a 3D scene. Your task is to plan the next sequence of camera actions to explore the scene and gather information needed to answer a spatial reasoning question. Available actions: • move-forward <distance> meters (ONLY use 0.25m increments: 0.25, 0.5, 0.75 etc.) • turn-left <angle> degrees (ONLY use 9° increments: 9, 18, 27 etc.) • turn-right <angle> degrees (ONLY use 9° increments: 9, 18, 27 etc.) CRITICAL RULES: 1. Plan 1–3 actions. 2. Each action on its own line, format exactly: <code>action_type value</code> 3. DO NOT output any explanation, reasoning, or commentary — actions ONLY. 4. Even if uncertain, always output a plausible exploration sequence. Example output: <pre>turn-right 18 move-forward 0.50 turn-left 9</pre>

Table 13: Prompt for the Answerer (Direct Answer)

Prompt for the Answerer (Direct Answer)
System Prompt: You are an AI assistant designed to help with spatial reasoning in a 3D scene. You must analyze any provided images or observations and answer the question. User Prompt: Image 1: <Initial Image> [Optional] Image 2: <Additional Reference Image> Question: {question} Answer Choices: • {choice_1} • {choice_2} • {choice_3} ... (additional choices) Output the exact answer from the choices. Answer:

Table 14: Prompt for the Answerer (Post-Exploration Answer)

Prompt for the Answerer (Post-Exploration Answer)

System Prompt:

You are an AI assistant designed to help with spatial reasoning in a 3D scene. You must analyze any provided images or observations and answer the question.

User Prompt:

Image 1: <Initial Image>

Action 1: {action_sequence_step_1} <Generated Image 1>

Action 2: {action_sequence_step_2} <Generated Image 2>

...

Question: {question}

Answer Choices:

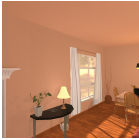
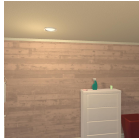
- {choice_1}
- {choice_2}
- {choice_3}

... (additional choices)

Output the exact answer from the choices.

Answer:

Table 15: Qualitative Examples of the SpaThor-1K.

Question & Answer	Initial View ($t = 0$)	Observation Sequence ($t = 1, 2, 3$)				Action Sequence
Q: how many burners are there on the gas stove ahead? A: One. B: Two. C: Four. D: Six.						[Move Forward 0.5m, Turn Left 18°, Move Forward 0.75m]
Q: How high is the refrigerator ahead? A: One meter eighty. B: One meter.						[Move Forward 0.75m, Move Forward 0.5m, Turn Right 36°]
Q: If I stand in front of the book on the table, where is the mirror? A: On my left. B: On my right.						[Move Forward 0.25m, Turn Left 9°, Move Forward 0.5m]
Q: What's the thing hidden behind the TV stand up ahead? A: A trash can. B: A carton.						[Move Forward 1m, Turn Left 27°, Move Forward 0.25m]
Q: Which is closer to the camera—the lamp or the window? A: The lamp. B: The window.						[Turn Right 36°, Move Forward 1m, Move Forward 0.75m]
Q: Starting from here, if I rotate 30° to the right, move forward 0.5 m, then rotate 30° right, would that put me facing the mirror? A: Yes. B: No.						[Turn Right 27°, Move Forward 0.5m, Turn Right 27°]